archaeometry

Archaeometry **60**, 1 (2018) 88–104 doi: 10.1111/arcm.12368

THE ARCHAEOMETRY AND ARCHAEOLOGY OF ANCIENT CHINESE GLASS: A REVIEW*

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This paper provides a new review of archaeometric research carried out on glass found in China, set in an archaeological context, from its earliest occurrence to the Song dynasty. It is set within a broad geographical context taking the terrestrial and maritime Silk Road contacts into account. We discuss chemical and isotopic compositional contrasts in glasses from different periods found in different parts of China, the glasses that were almost certainly made in China and those that were imported. A theme that runs through the paper is the problem of provenancing glass found in China, along with a lack of evidence for primary glass-making sites and minimal evidence for secondary glass production. We discuss the glass artefacts that are of typical Chinese types and contrast these with imports; the structure reflects this contrast. We discuss potential new scientific and archaeological approaches to Chinese glass.

KEYWORDS: GLASS, SILK ROAD, LEAD–BARIUM, POTASSIUM GLASS, PLANT-ASH GLASS, ROMAN, SASANIAN

INTRODUCTION

Glass first appeared in China at the earliest *c.*1000 BC, some 1000 years after its appearance in western Asia. The origins of Chinese glass were first addressed by Seligman and Beck (1938). Some of the beads that they examined differed compositionally from Western ones, containing high barium and lead, but were typologically similar. Needham (1962) and Caley (1962, 90) suggested that glass with high barium and lead levels found in China could have been made there because high levels of barium had never been found in ancient Western glasses. Even though high-lead glazes were made in the West as early as the first century BC (Walton and Tite 2010), Sayre and Smith (1961) and Needham (1962) pointed out that it is not until the eighth century that the first lead-silica glasses are found in the West.

Needham noted that the earliest Chinese high-barium glass may have emerged during the Spring and Autumn or the Warring states periods. Brill and his colleagues carried out a large number of scientific analyses of glass found in China in the 1970s and early 1980s (Brill *et al.* 1979, 1984), providing a starting point that is still widely used today; Warren (1977) published on Chinese glass vessels. During the 1980s and 1990s, research by senior scholars such as Jiayao

*Received 4 August 2017; accepted 15 November 2017
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An, Guangmei Shi, Boda Yang, and Fujia Gan and his team established a framework for the study of Chinese glass.

Fundamental publications (An Jiayao 1984) not only identified the presence of Roman, Sasanian and Islamic glass vessels in China, but related their presence to historical texts in which the vessels are mentioned. Five international conferences organized by Fuxi Gan and his team were important platforms for Chinese glass scholarship; the published proceedings have helped this research area to grow stronger. Chinese glass research has expanded even more since 2000. In this paper we aim to present an updated review of some important research areas on glass found in China, note some of the lacunae in our knowledge and suggest some possible ways forward.

The discussion below is set within a relatively broad geographical context and structured accordingly. The review covers the earliest glass to that dating to the Song dynasty. Little research has been carried out on Yuan, Ming and Qing dynasty glass, so it is considered premature to review it. The geological area covered by this review is basically the terrain of modern China. Prior the Western Han dynasty, Xinjiang was not considered Chinese because it lay to the west of the current Gansu Province. Therefore, variations in social and cultural contexts for the glass are also involved.

EARLY IMPORTED GLASS—AND FAIENCE—FOUND IN CHINA

It has been argued that the Silk Roads became evident during the Shang dynasty; it is generally agreed that the network of routes linking western Asia to China that formed the Silk Roads become more easily recognizable by the Western Han dynasty. Some of the earliest glasses found in China date to the Spring and Autumn period (770–476 BC). The inscription on a bronze sword, inlaid with three pieces of blue glass, from Wangshan Tomb 1, Hubei Province, states that the owner was King Goujian of the Yue State (*c*.520–465 BC) (see Fig. 1). Another, similarly inscribed, and decorated with colourless glass, from Henan Province belonged to Fuchai, the King of the Wu State (*c*.528–473 BC): this glass is of a soda–lime composition of a probable Western origin (see below).

Eleven glass eye beads from Hougudun Tomb I, Gushi, Henan Province (Henan Provincial Institute of Cultural Relics and Archaeology 2004), found in a tomb, were associated with an

Figure 1 *Glass decoration on the sword belonging to King Goujian of the Yue State (c.520–465 BC). [Colour figure can be viewed at wileyonlinelibrary.com]*

inscribed bronze vessel, which indicates that the occupant was the sister of the King of the Song State, who was married to Prince Fuchai of the Wu State; it dates to the end of the Spring and Autumn period (770–476 BC) (see Fig. 2)—one bead that has been analysed is of a soda–lime composition and therefore of a probable West Asian origin. Such beads have been found in tombs dating to the end of the Spring and Autumn period in Taiyuan, Shanxi and Luoyang, Henan. Somewhat earlier glass beads have been found in Xinjiang cemeteries, such as at Kizil, Xinjiang, and are of a definite western Asian composition (Li *et al*. 2014) in an area occupied by nomadic clans. Some clans were already trading with the Chinese—especially in jade, which was a crucial part of all ritual activities. The presence of Western glass could potentially have been one of the factors that stimulated the production of the first Chinese glasses.

Early glass found in China is mainly in the form of beads and jewellery, rather than vessels. The development of ceramics, especially glazed ceramics, received far more cultural investment than glass technology, providing coloured glazed vessels in the Warring States and Han periods (476 BC – AD 25) in place of coloured glass vessels. Jade was used widely in South-East Asia from as early as 3000 BC during the Neolithic to make a range of objects, including vessels (Braghin 2002; Hung *et al*. 2007; Gan *et al*. 2009); jade and glass are both translucent materials (Brill *et al*. 1991, 34; Gan 2005).

Semi-vitreous faience ('Liaozhu') was used as jewellery from the early Western Zhou to the Spring and Autumn period (1046–476 BC); its occurrence is later than in the West (Zhang 1991). Hundreds of faience beads deriving from five cemeteries in northern China (mainly Shaanxi) date to between 1000 and 768 BC. Both soda-enriched and potassium-enriched vitreous phases have been found (Lei and Xia 2015); the earliest K_2O vitreous phase dates from the middle Western Zhou (10th century BC). Lei and Xia suggest that the soda-enriched faience was made in Mesopotamia or between Mesopotamia and China (*ibid*. 41).

Faience beads from Chu tombs at Shenmingpu, Henan and Leigudun, Hubei dating to the middle to late Warring States period (475–221 BC) were analysed using ED-XRF (Yang *et al*. 2013). They contain 11.93–18.46% K_2O and <0.62% MgO, probably indicating that potassium nitrate (saltpetre) was used as a flux (Brill *et al*. 1991). Potassium-rich faience glazes (currently) predate potash glasses in China; it is likely that faience technology influenced and led to the

Figure 2 *Glass beads from Hougudun Tomb I, Gushi, Henan, dating to the end of the Spring and Autumn period (770–476 BC). The occupant was the sister of the king of the Song State, who was married to Prince Fuchai of the Wu State. [Colour figure can be viewed at wileyonlinelibrary.com]*

emergence of high-potassium glasses (Yang *et al*. 2013; Gu *et al*. 2014). It is also likely that this was an independent invention in China, although from the 10th century BC high-potassium glasses, also with low magnesia levels, were being made in the West, probably in the southern Mediterranean area (Henderson 2013, 196–7). A potential influence of Shang and Zhou protoporcelain glazes on the production of early Chinese glass, probably made with wood/plant ashes (Zhang 1985; Kerr and Wood 2004, 11; Yu *et al*. 2017), should also be taken into account. Nevertheless, by the Warring States and Eastern Han dynasties, lead and barium are found in both low-fired pottery glaze and glass beads, so in this case Chinese glass and glaze appear to have influenced one another, or such glass influenced the development of glazed ceramics (Wood 2001, 38–9, table 2).

EARLY IMPORTED GLASS BEADS FOUND IN XINJIANG

This western region consists of the lands west of the Yumen Pass and east of the Pamir Plateau, where trading between the West and the East passed along the early Silk Roads. Prior to the Western Han dynasty, this area was occupied by small nomadic nations, who took part in trade on the Silk Roads. Glass providing evidence of trade has been found on many sites dating to before the Han dynasty.

As noted above, examples of plant-ash glass beads assumed to have been imported from western Asia along the early Silk Roads and dating to the Spring and Autumn period of the Western Zhou dynasty (1046–771 BC) have been found in the Kizil cemetery, Xinjiang, in western China (Table 1; see also Li *et al*. 2014). Apart from two (later) high-potassium examples, the published results for unweathered Kizil cemetery beads are of a plant-ash soda–lime composition. The beads are mainly undecorated globular and annular types. One is dark blue with two white eyes (Li *et al*. 2014, fig. 2), a type that is found in Central and Western Europe (ninth to eighth centuries BC), at Must Farm, UK and Kaman, central Anatolia (both unpublished), so it may have originated in Europe or the Near East. In addition to soda, halophytic plants can introduce elevated levels of, for example, MgO, K_2O, P_2O_5, Cl, SO_3 and CaO, determined by, the genus/species of plants, the geological context in which the plants grow and the mixing of ashes of different compositions (Henderson 2013, 36–48). Kizil cemetery glasses are contemporary with glass from early first millennium Assyrian Nimrud, Iraq (Reade *et al*. 2005) and from Hasanlu, Iran (Stapleton 2003); there is increasing evidence for continuity in plant-ash glass technology from the second millennium BC into the first millennium BC (Henderson *et al*. in press). Therefore, it can be suggested that these Xinjiang plant-ash glasses had either a Mediterranean or a European origin, and that they were made sometime after *c*.1000 BC. Although is unlikely that they are (earlier) Late Bronze Age Egyptian or Mesopotamian glasses (cf., Li *et al*. 2014, 619), it is nevertheless curious that none of Kizil beads are of a natron composition, a dominant technology after *c*.800 BC in the West, with early examples dating to *c*.1000 BC. This may suggest that they derived from a limited geographical source.

The earliest natron glasses found in China are often eye beads found in high-status burials dating to the late Spring and Autumn to early Warring States periods, equivalent to the Phoenician period in western Asia (*c*.600–400 BC) (see Table 1 and Fig. 2). Examples are opaque yellow beads with stratified white and blue eyes, found also in Mediterranean contexts (Spaer 2001, 81–4). Two pale yellow glass vessels from the Bu-Gai-Wu-Yi-Li-Ke stupa in Xinjiang, dating to between the sixth and eighth centuries AD, are also of a natron composition (Liu *et al*. 2011, table 4, sample XJ-25).

Table 1 *The main compositional types of glass found in China (wt% oxide; TiO_2, Rb and Sr in ppm): PAG C As, (soda) plant-ash glass, Central Asia; PAG-'Sas', (soda) plant-ash glass, 'Sasanian'; High K, potassium oxide – silica glass; Pb–Ba, lead–barium glass; M alkali, mixed-alkali glass; Pb–Si, lead–silica glass; PAG W As, (soda) plant-ash glass, western Asia; K–Ca, potassium–calcium glass. Site names: qtm, Qiong-Ti-Mu, Xinjiang; dwkmu, Da-Wang-Ku-Mu, Xinjiang; spl, Shan-Pu-La cemetery, Xinjiang; rffc, Red Flag cemetery, Xinjiang; Kizil, Kizil cemetery, Xinjang; Hongsh, Hongshan cemetery, Jiangsu*

	PAG C AS	*PAG-'Sas'*	*High K*	*Pb–Ba*	*Natron*	*M alkali*	*Pb–Si*	*PAG W As*	*K–Ca*
	1	*2*	*3*	*4*	*5*	*6*	*7*	*8*	*9*
Na_2O	13.7	15	2.3	3.4	13.2	6.9	1.1	20.25	1.41
MgO	1.9	5.2	0.7	2.3	0.3	1.3	0.4	5.77	1.12
Al_2O_3	4.4	1.5	2.4	0.3	1.6	2.9	nd	0.69	3.23
SiO_2	55.9	66.3	73.5	6.6	75.3	74.7	33.7	60.25	72.04
P_2O_5	nd	nd	0.8	nd	nd	1.3	nd	0	0.89
Cl	1.4	0.9	0.5	2.3	1.5	1.3	0.3	0.99	0
K_2O	4	3.1	14.4	nd	1.3	8.7	2.3	2.81	12.54
CaO	4.4	7	3.2	1	4.2	2.6	0.5	6.8	4.48
TiO_2	1683	133	527	nd	369	905	nd	300	1900
MnO	0.1	0.1	nd	nd	1.7	nd	nd	0.08	0.13
Fe_2O_3	9.3	1	0.5	0.2	1	0.8	0.1	0.67	1.1
CoO	nd	nd	nd	nd	nd	nd	nd	0	0
CuO	nd	nd	1.6	0.2	nd	nd	0.1	1.18	2.87
Ba	nd	nd	nd	20.49	nd	nd	nd	0	0
Rb	110	48	12	29	38	96	29	0	106
SnO_2	0.5	nd	nd	nd	nd	nd	nd	0	0.026
Sb_2O_3	0	0	0	0	0	0	0	0.22	0
PbO	nd	nd	nd	21.5	nd	nd	61.6	0.17	0.022
No.	xjo5-3-8	xj-16-3	xj05-6-5	xj05-6-8	xj-7	xj05-5-1	xj-16-9	xj-1A	JS-TB-1
Site	Niya	qtm	dwkmu	dwkmu	spl	rffc	qtm	Kizil	Hongsh
Date*	2 BC – 4 AD	6-8 AD	2 BC – 9 AD	2 BC – 9 AD	3-4 AD	1 BC – 1 AD	6-8 AD	11–7 BC	473–468 BC
Colour	Black	Light green	Green–blue	Green–blue	Colourless	Aqua	Green	Blue	Blue
Artefact	Bead	Bead	Bead	Bead	Bead	Bead	Bead	Bead	Bead
Publication†	1	1	1	1	1	1	1	2	3

nd, Not detected; 0, not reported.

*In centuries, except for the final column, which reports an actual date range.

†Publications: 1, Liu *et al.* (2012); 2, Li *et al.* (2014); 3, Liu *et al.* (2015).

Abdurazskov (2009) and Brill (1972, 2001a, 2009) noted that many Central Asian glasses from the Uzbek sites of Bukhara, Tashkent, Samarkand and Fergana contained a minimum of 4% K_2O: western Asian glasses invariably contain less than 4% K_2O (Henderson *et al*. 2016, app. B). Second century BC to second century AD beads from Niya, south of the Taklamakan desert, mainly have Central Asian plant-ash compositions, with 4.1–5.4% K_2O, 1.9–3.8% MgO, 2.2–9.5% Al_2O_3 (see Table 1 and Fig. 3) and 6.0–8.1% CaO. Liu *et al*. (2012) noted that plant-ash glasses from the Ferghana valley, Uzbekistan (Osório 2005; Abdurazskov 2009) have high alumina but lower potassium oxide levels, possibly representing a subtype made in Uzbekistan. One possible explanation is that the plants used to make the Uzbek glasses grew on a contrasting geology to those used to make other Central Asian glasses (see above). For example, beads from Xinjiang analysed by Liu *et al*. (2012), mainly dating to before the fourth century AD, are Central Asian plant-ash glasses.

GLASS VESSELS IMPORTED TO CHINA FROM THE WEST

After Zhang Qian's envoy had contact with Central Asia during the second century BC, communication along early Silk Road networks during the Han dynasty became stronger, and direct contact between the East and the West became more intensive. Following the invention of glass-blowing in the first century BC, probably in the Levantine region, Roman glass vessels started to arrive in China. Ancient sources, including Greek ones, refer to the transport of glass across the Mediterranean to the East; a Chinese account of glass made in the Roman Empire states that it was of 10 colours (An Jiayao 1984).

Roman glass vessels have been found in tombs all over China, from Xinjiang to the southern coast and to as far north-east as Liaoning Province. Typical Roman wares include mosaic and pillar-moulded glass bowls (An Jiayao 1984). Preliminary chemical analyses show that these vessels are (natron) soda–lime glass with typical Roman glass chemical compositions (An Jiayao 2016).

Later plant-ash glasses found in China, dating to the Wei, Jin and Northern and Southern dynasties (AD 220–589), are contemporary with Sasanian glass (see Table 1; see also Brill 2005; Mirti *et al*. 2008, 2009). Historical sources refer to Western glass wares being a means

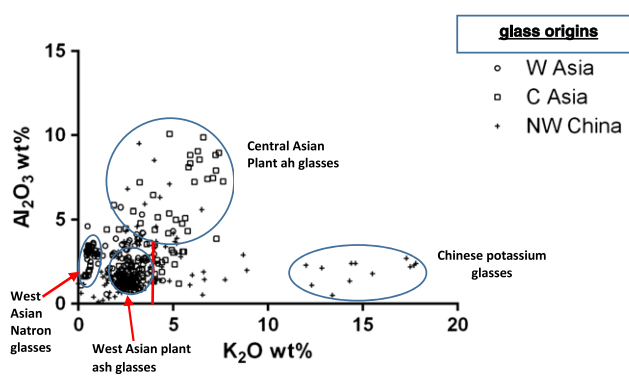

Figure 3 *The wt% Al_2O_3 versus K_2O in glasses found in western Asia, Central Asia and North-West China (data from Henderson et al. 2004 [Raqqa, Syria]; Brill 2009 [Loulan, Xinjiang and Panjikent, Tajikistan]; Rehren et al. 2010 [Uzbekistan]). [Colour figure can be viewed at wileyonlinelibrary.com]*

of showing off the wealth of their owners (An Jiayao 1984). Even more Sasanian than Roman glass vessels have been found in high-status tombs all over China. The typical wares include facet-cut bowls such as those found in the Western Jin period Huafang Tomb and the Northern Zhou period Lixian Tomb. Preliminary chemical analyses show that these wares are of soda–lime glass with typical Sasanian chemical compositions (An Jiayao 2016).

Firmly dated mainly second century BC to 10th century AD plant-ash glasses derive from a range of sites, such as Qiong-Ti-Mu and Da-Ma gorge, either north or south of the Taklamakan desert. These are referred to by Liu *et al*. (2012, 2137) as v-Na–Ca SAS, because they are similar in composition to Sasanian (plant-ash) glasses with K_2O and Al_2O_3 levels <4%; they include examples that predate the Sasanian period. Analyses of Iron Age glass beads in Taiwan of first millennium AD date have also revealed the presence of plant-ash glasses there (Wang and Jackson 2014).

Islamic glass vessels found in China include a glass bottle and a glass vase from the Kang Mausoleum of the Southern Han dynasty (AD 942) in Guangzhou. Fragments of Islamic glass vessels were found in the Attendant Tomb No. 1 in a mausoleum in Zuling, Inner Mongolia, dating to the Liao dynasty (AD 907–1125); one vessel is possibly from Nishapur. Some exceptional examples of late Tang dynasty plant-ash glass vessels are those found during excavations of the Famen temple near Chang'an/Xi'an. A wide range of artefact types were discovered, including silk fabrics, gold and silver vessels, pearls, pieces of jade and 19 porcelain vessels. More than 20 early Islamic glass vessels were found (Carboni 2001, 162), consisting of scratch-decorated, colourless cut, lustre-decorated and translucent cobalt blue examples (Li *et al*. 2016a, table 8.3). Low-vacuum p-XRF analysis of Famen vessels produced analysis totals close to 100%, with soda levels close to acceptable levels. All are plant-ash soda–lime glasses; the scratch-decorated vessels are of a possible Syrian origin and two colourless glasses are of possible Iranian or Iraqi origin (Li *et al*. 2016a).

LEAD–BARIUM GLASSES MADE IN CHINA

Soon after glass eye beads reached China, imitations appeared (Fig. 4); some beads from early Warring States period tombs look like Western glass eye beads. However, they are instead either glazed (some with Pb–Ba compositions) ceramic beads (Yang *et al*. 2013) or lead–barium faience beads. It has been argued that the lead–barium faience was possibly the precursor of the lead–barium glass, because they predate the glass by more than 100 years and they occur

Figure 4 *A lead–barium eye bead from tomb no. 16 of Pingliangtai Huaiyang County, Henan. [Colour figure can be viewed at wileyonlinelibrary.com]*

in the same area (Wang *et al*., due to be published in the same volume as this paper). Although later, it should also be noted that artificial barium-rich Chinese purple and blue pigments were produced in the late Warring States period (Qin *et al*. 2016). Thus, barium was present in faience, glazes, glasses and pigments.

High demand for glass impelled Chinese craftsmen to make glass eye beads. During the middle phase of the Warring States period, a large number of glass eye beads appeared in the burials of nobles and common people all over the country, but they are concentrated in Hunan, Henan and Hubei, and some are of a lead–barium composition. Perhaps because China had limited (mineral) soda resources, glassmakers had to find alternative raw materials and/or they built on an existing vitreous technology.

During the Han and Warring States periods, instead of producing glass vessels, white and green lead–barium glass was mainly used to make eye beads, tubular beads and earrings, ear and nose plugs, cicadas used in funeral deposits, ritual *Bi* discs, sword pommel decoration and large animals; in the Western Han dynasty, it was used to make burial suits. Sixteen moulded opaque green cups, amongst the earliest vessels from China, were found in the prince of Chu's tomb, Jiangsu, dating to 128 BC (An Jiayao 2004). Evidence of large-scale production of lead–barium glass is provided by a Western Han dynasty disc weighing 1.9 kg from the mausoleum of the Wudi, Liu Che (Gan 2005) and an estimated 22 glass musical instruments (chimes, Qing [磬]) from the Han dynasty Jiangdu king's mausoleum at Dayun Hill, Jiangsu, weighing 131 kg (Nanjing Museum and Bureau of Culture 2013; Li *et al*. 2016b) (see Fig. 5). A recent assessment of how lead–barium glasses were utilized yielded the following results: decoration and jewellery 85.5%, funeral goods 8% and ritual vessels 5.5% (Dong *et al*. 2015, 461).

Lead–barium glass was probably made in central China (Gan 2009, 25). The glass type has been found in North-West China (Xinjiang) and central China, in Jiangsu, Anhui, Shaanxi, Guangdong, Guangxi, Fujian and Chongqing Provinces, dating to the late Warring States, Han and Eastern Han dynasties (Beck and Seligman 1934, 982; Seligman *et al*. 1936; Gan *et al*. 1978, 99; Brill *et al*. 1991, 34–6; Gan 2009, table 1.6; Li *et al*. 2016b). The glass type also occurs in other parts of South-East Asia, such as in Japan and Korea (Lee 2009). Lead–barium glass typically contains 20.5–67.1% SiO_2, 12.3–54.5% PbO and 0.16–20.3% BaO. The total K_2O and Na_2O contents are generally ~2% (see Table 1; see also Dong *et al*. 2015). Higher PbO and BaO variations in Warring States than in Han dynasty samples could suggest that flux use changed between the middle Warring States and Eastern Han dynasties, a period of ~400 years, although the dating is not precise enough to be certain. Green and

Figure 5 *A lead–barium chime from a Han dynasty tomb at Dayun hill, Taiyu County, Jiangsu Province. [Colour figure can be viewed at wileyonlinelibrary.com]*

brown lead–barium glasses are mainly coloured by transition metal ions Cu^{2+} and Fe^{2+}. Nineteen out of the 22 chimes with unweathered compositions contain lead oxide levels of between 26.48 and 35.51% (Li *et al*. 2016b).

The mineral barite(s) is a possible barium source, but it is difficult to use in a glass melt. Although barium carbonate (witherite) is easier to melt, barite ($BaSO_4$) grains have been detected in samples using Raman spectroscopy (Dong *et al*. 2015, 465; Li *et al*. 2016b, 118). Raman spectroscopy has also provided evidence for feldspar family minerals, which may account for elevated Na_2O (max. 4.5%), K_2O (max. 4.5%) and Al_2O_3 (levels of 2.74–11.62% in a few lead–barium glasses; Dong *et al*. 2015, 465); Na_2O would be subject to leaching, so this may be a minimum level. Perhaps feldspar was added deliberately as a fluxing raw material. Some lead–barium glass samples contain elevated CaO of <6.6% (Cui *et al*. 2011; Li *et al*. 2016b). Perhaps these were made for a short time in the Warring States period: it appears that a calcium-rich raw material associated with Rb levels of ~100 ppm was used to make them. Cui *et al*. (2011) also noted Pb–Ba *Bi* discs and beads could be distinguished compositionally: the vessels contained >35% PbO and beads <30%. Reduced chemical compositions suggest that a soda–lime base glass, presumably of western or Central Asian origin, was used with added lead and barium. Recalculated compositions produce elevated Na_2O levels <9.61%, still rather low for a soda–lime glass, and only one contained sufficient calcium oxide (Brill 2001b, 450). Instead of the suggested introduction of feldspar minerals (see above), this might suggest that there was contact between workers of soda–lime and Pb–Ba glass and that the glasses/raw materials were mixed; more detailed scientific research of well-dated archaeological samples using more sensitive techniques is necessary to investigate this further.

Large numbers of Warring States lead–barium glasses occur in the Chu kingdom in south-east China, mainly in Hunan and Hubei (Hou 1995), which may have been a production zone. The technology may have been transferred from central China. In Jiangsu Province in eastern China, Western Han princes were buried with lead–barium glass *Bi* discs and chimes discussed above, and this may represent a second production zone (Dong *et al*. 2015). Some Chinese lead isotope signatures either have very high or very low lead isotope values (Brill and Shirahata 2009, figs 4.1–4.3); results for 12 lead–barium glass *Bi* discs and eye beads from Chu tombs in Hunan Province could suggest that they were made in Hunan Province (Cui *et al*. 2011). No primary or secondary glass production sites for lead–barium glasses have been recorded, so their exact provenance is still an open question.

Chinese lead–silica glasses contain a minimum lead oxide level of 35%, with other examples containing between 50% and 70% (see Table 1; see also Gan 2005, table 14.10; Gan 2009, table 1.8; Liu *et al*. 2012, table 3). Lead–silica glass is basically of the same composition as that found in the West from *c*. AD 800. The earliest Chinese examples postdate the first century AD; other examples date to the Tang dynasty (AD 618–907) (Gan 2009, table 1.8; Liu *et al*. 2012, table 3). Lead–silica glass has been found in contexts dating to the Kofun and Nara periods in Japan (AD 250–538 and 710–794, respectively); it continued to be used there in the 11th century (Inoue *et al*. 2016). Most contain between ~2% and 4.5% K_2O and a positive correlation between CaO and P_2O_5 could suggest that bone ash was used as a raw material. Early second millennium AD glass beads from Luliao, Taiwan are lead– silica or potassium–silica glass (Cui *et al*. 2008). Some AD 600–1200 Chinese high-lead glasses also contain potassium oxide or soda as well as silica with low or negligible calcium oxide (Gan 2005, table 14.10). Therefore, there is no doubt that the earliest lead–silica glass came from South-East Asia and only appeared in the West around AD 800, presumably as a result of technology transfer along the Silk Roads. Where in South-East Asia it was made is still to be confirmed.

Apart from lead–barium glass, soda–lime glass may have also been made in North China, perhaps from halophytic plants that grow there (Wang *et al*. 2015). A chapter on the Yuezhi people in *Bei shi* (*History of the Northern dynasty*) states that at the time of Emperor Taiwu of the Northern Wei (AD 424–52), Yuezhi people trading in Dai made glasses of various colours. Yuezhi people originated in the Kushan Empire, and Dai was the capital of Northern Wei, modern-day Datong in Shanxi Province. Preliminary analyses of glass from Northern Wei burials contemporary with the foundation of the Northern Wei pagoda in Datong, using ED-XRF at the University of Science and Technology Beijing, indicate that the glasses have compositional similarities to Central Asian glasses but that there are also differences; they therefore differ compositionally from Roman and Sasanian glasses that have been found in Northern Wei tombs. Because of their chemistry and because their shapes are the same as in contemporary Datong ceramics, it has been suggested that these vessels were produced using glass made by the Yuezhi people (An Jiayao 2016); research is ongoing. The first blown glass vessels were probably produced in China during the fifth century Northern Wei dynasty (An Jiayao 2004).

GLASS PROBABLY MADE IN SOUTHERN CHINA AND GLASS IMPORTED FROM SOUTH-EAST ASIA

A large number of glass beads, ornaments, burial objects and vessels have been found in South-West China, especially in Han-dynasty contexts in Guangxi, Guangdong, Yunan, Hunan and Gansu. K–Si and K–Ca–Si glasses have been found in Han dynasty tombs in Guangxi, Guandong, Yunan, Hunan, Gansu and Jiangsu. There may be much later (8th–12th centuries AD) Chinese examples of K glasses (Brill *et al*. 1991, table 2, no. 5805), but they are not closely dated. Of 136 chemical analyses of glass objects from Guangxi, one eighth are lead–barium ritual *Bi* discs and burial objects, while two thirds, mostly glass beads, are high-potassium. K–Si glass is also found at sites in South Asia, South-East Asia, Korea and Japan (Lee 2009, 186).

High-potassium glasses are rarely found in western Asia until *c*. AD 800: there are, however, some Late Bronze Age examples of 11th–10th century BC date (Henderson 2013, 196–9). Chinese examples are generally characterized by >15% K_2O, negligible Na_2O (<1%), low MgO (<1%), ~2% CaO or less, variable Al_2O_3 (0.8–6.5%) and relatively high Fe_2O_3 (<3%), suggesting the use of impure sands; variable Al_2O_3 and CaO suggest that a variety of sand sources were exploited (see Table 1). High CaO levels probably indicate that a Ca-rich raw material was used. The low Na_2O and MgO levels indicate that a mineral source of potassium, saltpetre (KNO_3), was used in the glass. The environment and climate in Guangxi and Guangdong in south-east China, with sufficiently high temperatures and suitable humidity, allow for the formation of saltpetre, an efflorescent salt that would have been purified by leaching and recrystallization (Brill *et al*. 1991). Saltpetre was used as a medicine during the Western Han period (Zhao 1991): its availability and processing may have provided the basis for glass production, although this does not constitute proof. Gan (2005) published results for Chinese K–Ca–Si glasses with 6.31–11.36% CaO. It is not until the 14th century that Chinese (low-fired) pottery glazes incorporated high levels of potassium (Wood 1999, 217, Table 92), so in this period the development of glass and glaze followed distinctive and separate developmental pathways.

K–Si glasses are found in South and South-East Asia, including China, Japan and Korea (Lee 2009, 186). This therefore raises the issue of where the glass was made and whether it was made in multiple centres. One of the earliest examples of a K–Ca–Si glass is a sword ornament found in the tomb of Guo Jian, prince of Yue (496–464 BC) in Hubei (Hou 1991): the Yue peoples of southern China and northern Vietnam may have been involved in its initial manufacture (Lankton and Dussubieux 2013, 418). Liu *et al*. (2015) report early Warring States beads from Yue state

burials (473–468 BC) with potash–lime compositions (Table 1). While it is highly probable that glass containing high potassium levels was made in south-east China, where saltpetre occurs, Li *et al*. (2014) have suggested that it was made in South-East Asia and South Asia (India) too.

Three compositional subgroups of high-potassium glasses have been identified (Lankton and Dussubieux 2006; Dussubieux *et al*. 2010; Dussubieux 2016, table 5.2):

a m-K–Ca–Al, with moderate levels of K and Ca (below 3%) and with Al_2O_3 levels of between 1 and 4%;
b m-K–Al, with high alumina levels (3–7%) and low CaO levels (at ~1–3% and below);
c m-K–Ca, with low Al_2O_3 (~3% or below) and high CaO (3–7%).

Type (a) is found more frequently in southern Asia—and especially at Arikamedu in India (though poorly dated)—than in South-East Asia, Vietnam and southern China. Type (b) may have been manufactured in southern China (Lankton and Dussubieux 2013), but it is also found in southern Cambodia, northern Vietnam and Thailand. Moulded bowls with ribs dating to the Han dynasty, mainly found in South-West China and Vietnam, were made with this type of glass. Their manufacture in this area is supported by historical evidence (Borell 2016, 62), although this is not evidence of primary glass-making. Seven sixth to eighth century vessel fragments from Qiong-Ti-Mu, Xinjiang and three other Tang dynasty fragments from Bo-Xi-Ke-Re, Xinjiang (Liu *et al*. 2011, table 4) are characterized by <1% MgO, 5.0–7.99% Al_2O, 7.5–9.0% K_2O and 1.8–3.5% CaO. They also contain ~13–18.9% Na_2O. Type (c) has been found especially in Thailand, at Ban Don Ta Phet. It may have been made in China or South-East Asia (Glover and Henderson 1995), including Vietnam (Lankton and Dussubieux 2006).

Ten fourth century BC to fourth century AD glass beads from Xinjiang of the m-K–Ca–Al type contain 0.5–2.4% Al_2O_3, 1.4–4.2% CaO, 12–17.8% K_2O and 0.5–0.9% MgO (Liu *et al*. 2012, table 3). Furthermore, four early Eastern Han high-potassium glasses from Feng-ma-ling are characterized by low MgO levels of <0.7%, 3–9% Al_2O_3 and 0.4–0.8% CaO (one contains 4.2% CaO) (Li *et al*. 2009, tables 2.15 and 21.6), making them indistinguishable from the early European (early Iron Age) examples (Henderson 2013).

The Rb:Sr ratio in some m-K–Al is between 10 and 30; the ratio in m-K–Ca–Al is usually <4 (Xiong and Li 2011). There are notable exceptions, such as amongst fourth to second century glasses from Khao Sam Kaeo and Ban Don Ta Phet in Thailand (Lankton and Dussubieux 2013, table 5.7.2). These elements are normally associated with lime, so the contrasting ratios probably reflect the use of different sources of lime, including geological ones. On the basis of Rb and Sr levels, Liu *et al*. (2013) have defined two compositional groups of Han dynasty K–Si glass beads from tombs in Hepu county, Guangxi: high-Sr, low-Rb and high-Rb and lower Sr. High-Rb, low-Sr is correlated with elevated Al_2O_3 in some cases. Correlated elements were also apparently introduced in K-rich minerals in the sands, such as potassium-rich feldspars, rather than solely with lime. Changing lime–strontium correlations in fourth century BC and second century AD high-potassium glasses from India and Thailand (Dussubieux 2016) show that lime sources changed over time.

Strontium isotope investigations of high-K glasses from China, Thailand, India, Korea, Vietnam, India and Japan by Brill and Stapleton (2012) reveal at least three different groups. These groups may, or may not, be related to the varying concentrations of rubidium and strontium discussed by Liu *et al*. (2013) and Dussubieux (2016). Nevertheless, it can be suggested tentatively that the results provide evidence for separate primary production zones for K–Si glasses.

Lead isotope analysis of K–Si glasses does not help to provenance them. Some blue glasses tested by Brill and Shirahata (2009, 156–8) are coloured by cobalt: the isotope results for 11 of them fall at an intermediate point between the extreme values for Chinese glasses (see above)

and probably reflect the isotope values of the cobalt-rich mineral employed. No primary or secondary glass production sites have been found for potassium glasses.

Three 25 BC – AD 220 mixed-alkali green glass beads from Feng-men-ling, Guangxi, analysed using ICP–AES, contain >6% Na_2O and K_2O, 1.9–10.9% PbO, ~1% CaO and <0.5% MgO (Li *et al*. 2009, table 21.5). However, the MgO levels in mixed-alkali glasses do not always suggest the use of a mineral alkali (Table 1). A further (first century BC to first century AD) example derived from the Red Flag Factory cemetery, Xinjiang (Liu *et al*. 2012, table 3).

Brill (1987) noted that high-alumina soda glass was a characteristic of South-East Asian glass and that the soda-rich mineral *reh* was probably used to make it (Brill 1999, vol. 2, 481; Dussubieux 2001, 139). It is characterized by high levels of Na and Al, variable Ca and low levels of Mg. The type can be split into five subtypes (Dussubieux *et al*. 2010, 1647) according to the levels of Ca, Mg, U, Zr, Ba, Cs and Sr. One or more of the five subtypes can be found not just in South-East Asia but also in South Asia (North and Western India, Sri Lanka), Turkey and East Africa. Primary production outside China is certain but it is impossible to be more precise. Indian-Pacific Ocean glass beads with high alumina were transferred to the central plains as Buddhism flourished in China. Examples are the 150 000 beads found in the Yongningsi temple, Luoyang, built by the Empress Hu in AD 516 (An 2000). Otherwise, such glass is currently surprisingly rare in China.

OPACITY

'Western' opacifiers have been identified in Chinese glass mainly by using SEM–EDS and Raman spectroscopy. Li *et al*. (2014) have identified lead antimonate ($Pb_2Sb_2O_7$) and calcium antimonate ($Ca_2Sb_2O_7$) in beads from the cemeteries at Kizil and at Wanquan, Xinjiang. The glass seems to be the earliest found in China, dating from the Western Zhou to the Spring and Autumn period (*c*.1000–600 BC); their matrices are soda–lime plant-ash compositions and are of a probable Western origin (see above). The two bluish-green 500–300 BC K_2O beads from the Kizil cemetry contained cassiterite (Li *et al*. 2014, table 1, fig. 7). Given tin oxide levels of 6.56% and 5.83%, it is likely that they are opacified with tin oxide, making them an early example of tin opacification in a non-Western glass (though it is not stated). First century AD tin-opacified white and yellow glasses from Guangxi Zhuang have been identified using confocal Raman microspectroscopy (Zhao *et al*. 2013). Tin-opacified first century BC to 10th century AD glass beads also derive from Xinjiang and Guangxi (Li *et al*. 2013). The first century BC date accords with a peak in tin opacification in prehistoric Europe (Henderson 2000, 36); there is, however, evidence that tin opacification was used as early as the sixth century BC in Eastern Europe (Purowski *et al*. 2012). The use of tin opacification in Chinese high-potassium glasses hints potentially at an 'opacifying' independent invention.

There are 'opacifying' materials that are unique to South-East Asia. By the Tang dynasty (AD 618–907), calcium fluoride (fluorite, CaF_2) was used to produce an *opalizing* effect in imitation of jade (Werner and Bimson 1963, 303). Turbidity (not the same as opacity) has been observed in soda–lead–barium glass (Brill *et al*. 1991, 34, fig. 19); it is probably due to (localized) devitrification and was probably used to imitate jade. Recently, CaF_2 has been found in potassium–lime glass attached to the inner wall of a Yuan dynasty (AD 1115–1368) glass-working or -making crucible found at Yanshenzhen, Shandong Province (Professor Cui Jianfeng pers. comm.). Somewhat later, it was used as an opacifier in 16th and 17th century Chinese cloisonné enamels (Henderson *et al*. 1989).

CONCLUSIONS

1 In spite of high concentrations of K and Pb–Ba glass in China, a lack of archaeologically excavated primary glass production sites with furnaces makes it difficult to be sure where exactly such glasses were made and therefore to be able to provide a provenance for excavated glass artefacts of these compositions. Absolute archaeological proof should include semi-fused glass but no such proof has been found in China. Nevertheless, it is very likely that both K and Pb–Ba glasses were made in China.

2 There is little apparent evidence for glass-working in China, including the discovery and export of raw furnace glass and glass ingots, crucibles in which glass was reheated so as to work it or even glass-working furnaces. The Yanshenzhen crucible noted above is an exception.

3 New analytical approaches to glass found in China may provide a way forward. Neodymium and strontium isotope analyses may help to provenance it, although even here, without establishing the isotopic characteristics of silica deposits and Sr characteristics of calcium-bearing raw materials across the landscape, this will not necessarily provide a *final* provenance. As with its application to glasses (Henderson *et al*. 2005), strontium isotope analysis (together with strontium concentrations) will provide insights into types of lime sources used and therefore help to define the technology used: this should reflect the contrasts in Rb concentrations found in some K glasses. If Sr and Nd isotope analyses are combined in an environmental context, there is a potential to provide provenance information. Greater use of LA-ICP–MS, a microdestructive technique, to analyse Chinese glass, has the potential to define clearer compositional groups, and will lead to a clearer understanding of the raw materials used and potentially provide provenance for glasses made in China, including (subgroups of) high-K glasses, K–Ca glasses, Pb–Ba glasses and Pb–Ba glasses with elevated Na and Ca levels.

4 Early plant-ash glasses were made in Central Asia. However, where the environment is appropriate, such as in semi-deserts in which halophytic plants grow—as, for example, in North-West China (Wang *et al*. 2015)—there is at least the potential that plant-ash glass was made in such a location.

5 High-alumina soda glass is one of the commonest glass chemical compositions found in South-East Asia in the period under consideration. However, currently this does not seem to apply to China, where high-potassium glasses are more common. It might be suggested that the distribution of high-alumina glasses was tied to the maritime Silk Road and that inland locations in China had less access to such glass in spite of an inland network of canals. Therefore, this observation still needs to be explained fully.

6 There is intriguing evidence for links along the Silk Roads: West Asian eye beads, which have been found in Western Zhou contexts in North-West China, were copied/reinvented using Pb–Ba glass. Throughout the first millennium AD, links along the land-based and maritime Silk Roads (Krahl 2011) have led to the import of Roman, Sasanian and Islamic glass vessels, usually deposited in high-status contexts. These interactions can be defined more closely with a larger programme of analyses of well-dated glasses.

7 Whether Chinese glasses were private possessions, ritual objects or traded commodities, it is important to attempt to reconstruct the *chaîne opératoire* involved in the production of glass, and its potentially forging links between the production and use of glass, its relative values and the various social, economic and ritual contexts.

8 When new glass technologies are introduced, there is often a period when transitional, sometimes mixed, compositions can be identified (e.g., Henderson 2013, 97–102). The elevated

soda and lime levels found in some lead–barium glasses may be evidence of this. It would be worth investigating whether there is evidence for mixed glasses during other transitional periods.

9 The compositional similarity between early high-potassium glass found in China and the (earlier) Late Bronze Age glass found in Italy and Greece needs to be investigated comprehensively so as to ascertain whether the two are connected in some way.

10 There is potential to investigate further the technological links between vitreous technologies such as between ceramic glazes, faience, glass and enamels.

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